RideTrack
Live bus and train tracking with QR ticketing for Sri Lanka

Group Y3.S2.WE_62
IT3060 Human Computer Interaction
Sri Lanka Institute of Information Technology (SLIIT)

FAX

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To	The Director / Chairman
Organisation	Department of Transportation (Bus Services)
Attention	Operations / Passenger Services
Fax number	_____
From	Silva D S J, Team Leader, RideTrack project team
Organisation	Faculty of Computing, SLIIT, Malabe
Sender contact	**+94 70 417 4576** (Silva D S J)
Date	13 October 2026
Subject	**RideTrack mobile app: project brief and request for feedback and guidance**

Action requested

☐ Urgent ☐ For review ☐ Please comment ☐ Please reply

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2	Covering letter
3	RideTrack project brief (latest version, 13 October 2026)
4	Reply slip (please complete and fax back)

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13 October 2026

The Director / Chairman
Department of Transportation (Bus Services)

Dear Sir / Madam,

RideTrack: a mobile app for live bus tracking and QR ticketing

We are a team of four third-year undergraduates at the Sri Lanka Institute of Information Technology (SLIIT). As part of the IT3060 Human Computer Interaction module we are building **RideTrack**, an Android app that shows passengers where buses and trains are, when they will arrive, and lets them buy a QR ticket that a conductor or ticket inspector scans on board.

Bus passengers often wait at stops without knowing when the next bus will arrive, and cash ticketing slows boarding. RideTrack aims to give passengers live arrival information and a QR ticket that a conductor validates with a phone camera, while giving the authority a live view of services and a direct channel for service alerts.

The app includes a dedicated **transport authority** section in which officers manage vehicles, routes, stops, schedules and fares, watch live operations, review incidents reported by staff, and publish service alerts that reach passengers' phones immediately. We have designed this section with the department officers in mind and would value your view on whether it reflects how your work is actually done.

We respectfully request:

- Feedback on the authority and conductor workflows in the app.
- Guidance on using publicly available bus route, stop and fare information in the app for demonstration.
- A contact officer, and if convenient, a short demonstration to your team.

A one-page brief describing the app and its current status is attached, followed by a short reply slip. RideTrack is an academic project. It is not affiliated with or endorsed by your department, handles no real payments, and will not be released to the public under your name without your written consent.

Thank you for your time and consideration.

Yours faithfully,

Silva D S J (IT23652200)
Team Leader, RideTrack
Contact: +94 70 417 4576
on behalf of Group Y3.S2.WE_62, SLIIT

Team members: Silva D S J (IT23652200), Rajapaksha M.P.K (IT23665620), Herath H.M.S.G (IT23634626), Fernando H L R D (IT23635302).

RideTrack: Project Brief

What it is

An Android app for Sri Lankan public transport. A passenger first chooses **bus or train**; the whole app then shows only that type. Three user roles share one live database:

Role	What they can do
Passenger	Search routes, filter and sort trips, see stops and facilities, track a vehicle live on a map, book and pay (simulated), keep a QR ticket, cancel, view history, receive alerts.
Conductor / inspector	Sign in as staff, view the assigned journey, scan and validate QR tickets, see passenger counts, report incidents and delays, complete the journey.
Transport authority	Manage vehicles, routes, stops, schedules, fares and facilities; preview and publish changes; watch live operations; review incidents; publish alerts.

Links between roles: (1) the passenger's QR ticket is scanned by the conductor; (2) a conductor's incident report reaches the authority; (3) an authority alert appears in passengers' notifications.

Current status

Area	Status
Design (requirements, prototypes, user testing)	Complete: 42 screens designed and tested with users
Passenger search and live tracking (8 screens)	Built. Home reads live data; other screens use sample timetables. Vehicle movement is simulated from the timetable.
Booking and QR tickets (11 screens)	In development
Conductor / inspector (10 screens)	In development
Transport authority (13 screens)	In development; shared database and security rules prepared
Testing and release	Functional, integration and usability testing (5+ participants) to follow; Android APK release

Technology and data

- React Native with Expo (SDK 57), TypeScript, Google Maps; Firebase Firestore and Authentication for real-time data and role-based sign-in.
- Payments are simulated; no card or bank details are collected.
- All route, timetable and fare data in the app is currently sample data prepared by the team, not official data.
- Access to the database is controlled by role-based security rules; passenger data is used only for tickets and alerts.

How the department could benefit

- Route changes and disruption alerts published once, delivered to every affected passenger's phone.
- Conductors validate QR tickets with a phone camera; reused tickets are rejected.
- A live operations map and incident reports from conductors in one place.

Reply Slip

Please complete and fax back, or call Silva D S J on +94 70 417 4576

Organisation	Department of Transportation (Bus Services)
Responding officer	
Designation	
Phone / email	

Please tick all that apply

☐	We are willing to give feedback on the authority section of the app.
☐	We are willing to attend a short demonstration (in person or online).
☐	You may use publicly available bus routes, stops and fares in the app for demonstration.
☐	Please contact the officer named above for further discussion.
☐	We are not able to assist at this time.

Comments

Signature	
Date	
Official stamp	